⚠️ This website is currently undergoing data validation. Certain datasets are pending review and dependent on updates from the source data team responsible for compiling and delivering the information.

Acme United Corporation_10/10/2025

12 minute read

Warning Letter 320-26-03

October 10, 2025

Dear Mr. Johnsen:

The United States Food and Drug Administration (FDA) inspected your drug manufacturing facility, Acme United Corporation, FEI 3012129555, at 301 Marianne St., Shelton, CT, from March 18 to 28, 2025.

This warning letter summarizes significant violations of Current Good Manufacturing Practice (CGMP) regulations for finished pharmaceuticals. See Title 21 Code of Federal Regulations (CFR), parts 210 and 211 (21 CFR parts 210 and 211).

Because your methods, facilities, or controls for manufacturing, processing, packing, or holding do not conform to CGMP, your drug products are adulterated within the meaning of section 501(a)(2)(B) of the Federal Food, Drug, and Cosmetic Act (FD&C Act), 21 U.S.C. 351(a)(2)(B).

We reviewed your April 18, 2025 response to our Form FDA 483 in detail and acknowledge receipt of your subsequent correspondence.

During our inspection, our investigator observed specific violations including, but not limited to, the following.

1. Your firm failed to use equipment in the manufacture, processing, packing, or holding of drug products that is of appropriate design, adequate size, and suitably located to facilitate operations for its intended use and for its cleaning and maintenance. Your firm also failed to thoroughly investigate any unexplained discrepancy or failure of a batch or any of its components to meet any of its specifications, whether or not the batch has already been distributed (21 CFR 211.63 and 21 CFR 211.192).

Your firm used **(b)(4)** water as a component to manufacture your topical over-the-counter (OTC) drug products, such as antiseptic wipes, labeled as sterile and non-sterile. You failed to adequately design and control your **(b)(4)** water system to ensure that it was suitable for producing water used in the formulation of your drug products. For example, you isolated the objectionable microorganism *Burkholderia cepacia (B. cepacia)* on numerous occasions within the holding tank and at the point of use. You regularly sanitized your water system after recovering *B. cepacia*, yet you continued to recover *B. cepacia* shortly after those repeated sanitization cycles. Microbiological monitoring of your system also yielded several "too numerous to count" results at multiple points in the system. These data indicated the presence of a recalcitrant biofilm. Furthermore, your firm failed to conduct timely and comprehensive investigations into these out-of-limit (OOL) microbiological results, including documenting root cause determinations and product impact assessment.

We acknowledge you recently installed a new water system. We note that this system, based on discussions during our inspection, appears to be an on-demand system without continuous circulation.

In your response, you state you do not believe your products pose a safety risk to consumers. For your non-sterile drug products, you indicate that there is a low potential impact to the end user because little to no microbial contamination was recovered from batch samples. You also indicate the active ingredient has an anti-microbial effect. Your response is inadequate. Antimicrobial formulations are not a substitute for adequate CGMP. Additionally, it is important to note that microbial contamination is not uniformly distributed, and samples may not be representative of the type or level of contamination that may exist within an entire system.

Your new water system continues to have deficient elements that do not assure long-term system reliability and suitable water quality. The **(b)(4)** water system must be adequately designed and properly maintained to minimize the potential for contamination. For example, stagnant water from an on-demand, non-recirculating system can foster the development of biofilms. **(b)(4)** water must be suitable for its intended use and routinely tested to ensure ongoing conformance with appropriate chemical and microbiological attributes.

In response to this letter, provide:

- A comprehensive assessment of your water system design, control, and maintenance. Regarding system control, for example, explain whether your firm is utilizing a recirculating water system and whether heat is an element of the system. In addition, state the duration, if any, that your

water system is continuously circulating, and provide your rationale for any duration less than 24 hours.

- A thorough independent assessment to ensure you have installed and are operating a suitable water system, accompanied by a remediation plan to address any identified deficiencies. Also, include a robust ongoing control, maintenance, and monitoring program to ensure the new system consistently produces water adhering to **(b)(4)** Water, USP monograph specifications, and appropriate microbial limits.
- Regarding the latter, ensure that your total microbial count limits and objectionable microbiological standards for water are appropriate in view of the intended use of the products produced by your firm.
- An updated risk assessment addressing the potential effects of the observed water system failures on the quality of all drug product lots currently in U.S. distribution. Specify actions that you will take in response to the risk assessment, such as customer notifications and product recalls.
- An action plan and timelines for conducting full microbiological testing of retain samples of your non-sterile finished drug products to determine the quality of all batches of drug product distributed to the United States that are within expiry. You should test all appropriate quality attributes including, but not limited to, microbiological quality (total counts and identification of bioburden to detect any objectionable microbes) of each batch. If testing yields an OOL result, indicate the corrective actions you will take, including notifying customers and initiating recalls.
- A summary of all data from all (e.g., accelerated, annual) stability studies from the last five years. Include all lots, all timepoints, and any out-of-specification (OOS) results, irrespective of whether they were later invalidated.
- A summary table of all individual microbial test results (total counts and identification of bioburden to detect any objectionable microbes) obtained from testing samples from your new water system. Also, include all physical and chemical test results (total organic carbon, conductivity, etc.)
- A retrospective review of all invalidated OOL water test results, including the following for each OOL:
  • For all OOL results found by the retrospective review to have an inconclusive or no root cause identified in the laboratory, include a thorough review of production (e.g., batch manufacturing records, adequacy of the manufacturing steps, suitability of equipment/facilities, variability of raw materials, process capability, deviation history, complaint history, batch failure history). Provide a summary of potential manufacturing root causes for each investigation, and any manufacturing operation improvements.
- A comprehensive review and remediation plan for your OOS/OOL result investigation systems. The corrective action and preventative action (CAPA) should include but not be limited to addressing the following:
  • Quality unit oversight of laboratory investigations

- Identification of adverse laboratory control trends
- Resolution of causes of laboratory variation
- Initiation of thorough investigations of potential manufacturing causes whenever a laboratory cause cannot be conclusively identified
- Adequately scoping of each investigation and its CAPA
- Revised OOS/OOL investigation procedures with these and other remediations

2. Your firm failed to establish written procedures for production and process control designed to assure that the drug products you manufacture have the identity, strength, quality, and purity they purport or are represented to possess. Your firm also failed to follow appropriate written procedures that are designed to prevent microbiological contamination of drug products purporting to be sterile, and that include validation of all sterilization processes (21 CFR 211.100(a) and 21 CFR 211.113(b)).

You failed to adequately validate your production and process controls used in the manufacture of your drug products. Additionally, you failed to conduct adequate qualification activities for equipment used to manufacture your drug products.

Furthermore, your firm failed to follow appropriate written procedures that are designed to prevent microbiological contamination of drug products purporting to be sterile, and that included validation of all sterilization processes. For example, your firm has not performed quarterly dose audit studies to monitor the **(b)(4)** dose to ensure that it remains effective for the **(b)(4)** sterilization of your drug products intended to be sterile.

In your response, you state that you will develop and implement enhanced process controls and complete process validation studies. Your response is inadequate as it did not provide a timeframe for completion of process validation activities for each of your drug products. In addition, your response did not include your interim plan for any drugs distributed before validation activities are completed to ensure you produce drug products of acceptable quality.

Process validation evaluates the soundness of design and state of control of a process throughout its lifecycle. Each significant stage of a manufacturing process must be designed appropriately and assure the quality of raw material inputs, in-process materials, and finished drugs. Process qualification studies include intensive monitoring and testing throughout each significant process stage to characterize intra-batch variation and evaluate batches to determine whether an initial state of control has been established.

Successful process qualification studies are necessary before commercial distribution. Thereafter, ongoing vigilant oversight of process performance and product quality is necessary to ensure you maintain a stable manufacturing operation throughout the product lifecycle. See FDA's guidance for industry, *Process Validation: General Principles and Practices*, for general principles and approaches that the FDA considers appropriate elements of process validation at https://www.fda.gov/regulatory-information/search-fda-guidance-documents/process-validation-general-principles-and-practices.

In response to this letter, provide:

- A detailed summary of your validation program for ensuring a state of control throughout the product lifecycle, along with associated procedures. Describe your program for process performance qualification, and ongoing monitoring of both intra-batch and inter-batch variation to ensure a continuing state of control.
- A timeline for performing process performance qualification for each of your marketed drug products. Also provide a risk assessment and any follow-up actions to be taken for the distributed drug products produced without performing any process validation studies.
- A detailed program for designing, validating, maintaining, controlling and monitoring each of your manufacturing processes that includes vigilant monitoring of intra-batch and inter-batch variation to ensure an ongoing state of control. As part of your response, explain how you handle changes throughout the lifecycle, including determining whether any changes (e.g., materials, manufacturing process, atypical bioburden) adversely impact the effectiveness of your **(b)(4)** sterilization process. Finally, include your program for qualification of your equipment and facility.
- A list of all quality complaints for the past five years. Include the batch number, name of product, lot size, whether a complaint sample was received (if not, explain why), and description of any associated CAPA.

3. Your firm failed to conduct appropriate laboratory testing to determine whether each batch of drug product purporting to be sterile conforms to such requirements (21 CFR 211.167(a)).

Your firm released your finished OTC drug products to the U.S. market without testing each batch for all appropriate quality attributes. For example, your firm distributed multiple batches of purportedly sterile finished drug products without conducting sterility testing. Without testing all appropriate physical and microbiological quality attributes of your drug products, your quality unit lacks essential safety information required to make suitable batch release decisions.

In your response, you acknowledge that you did not begin sterility testing on your drug products until the end of 2024. You indicate that you will test the retains, per USP <71> Sterility Tests, of lots for which you have retains.

Your response is inadequate because you have not provided proof of method suitability of the sterility test performed by your contract laboratory to ensure its capability to detect microbes in your finished drug product. Additionally, you have failed to indicate if there are batches for which you lack retains and are therefore unable to test for sterility.

In response to this letter, provide:

- A comprehensive, independent assessment of your laboratory practices, procedures, methods, documentation, supervision, and analyst competencies. Based on this review, provide a detailed

CAPA plan to remediate your laboratory system. If a contract facility performs any laboratory functions on your behalf, conduct an equivalent comprehensive assessment of their practices, implement an appropriate supplier CAPA, monitor the remediations at the contractor, and provide a summary of these planned CAPA activities.

- A summary of all test results obtained from testing your batch retains, including any OOS/OOL results obtained (irrespective of whether later invalidated). Also include a list of batches for which you have no retains and are unable to test for sterility.
- A list of microbial test methods and specifications used to analyze each batch of your drug product before making a batch disposition decision, and the associated written procedures.
- Method suitability tests performed for each of your formulations.

Quality Systems

Your firm's quality systems are inadequate. For guidance on establishing and maintaining CGMP-compliant quality systems, see these FDA guidance documents:

- *Q8(R2) Pharmaceutical Development* at https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q8r2-pharmaceutical-development
- *Q9 Quality Risk Management* at https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q9r1-quality-risk-management
- *Q10 Pharmaceutical Quality System* at https://www.fda.gov/regulatory-information/search-fda-guidance-documents/q10-pharmaceutical-quality-system

Conclusion

The violations cited in this letter are not intended to be an all-inclusive list of violations that exist at your facility. You are responsible for investigating and determining the causes of any violations and for preventing their recurrence or the occurrence of other violations.

Correct any violations promptly. Failure to promptly and adequately address this matter may result in regulatory or legal action without further notice including, without limitation, seizure and injunction. Unresolved violations may also prevent other Federal agencies from awarding contracts.

Failure to address violations also cause FDA to withhold issuance of Export Certificates. FDA may withhold approval of new applications or supplements listing your firm as a drug manufacturer until any violations are completely addressed and we confirm your compliance with CGMP. We may re-inspect to verify that you have completed corrective actions to address any violations.

This letter notifies you of our findings and provides you an opportunity to address the above deficiencies. After you receive this letter, respond to this office in writing within 15 working days. Specify what you have done to address any violations and to prevent their recurrence. In response to this letter, you may provide additional information for our consideration as we continue to assess

your activities and practices. If you cannot complete corrective actions within 15 working days, state your reasons for delay and your schedule for completion.

After you receive this letter, respond to this office in writing within 15 working days. In response to this letter, you may provide additional information for our consideration as we continue to assess your activities and practices, and/or submit a request to schedule an FDA inspection.

Send your electronic reply to CDER-OC-OMQ-Communications@fda.hhs.gov. Identify your response with FEI 3012129555 and ATTN: Nancy Scheraga.

Sincerely,
/S/

Francis Godwin
Director
Office of Manufacturing Quality
Office of Compliance
Center for Drug Evaluation and Research